higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

AUTOMOTIVE REPAIR AND MAINTENANCE
NQF LEVEL 3

11 MARCH 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1
1.1.1 D
1.1.2 E
1.1.3 C
1.1.4 B
1.1.5 F

(5 × 1) (5)

1.2
- Clamp the strut in an engineering vice without damaging it.
- Fit spring clamps and compress the coil springs evenly.
- Loosen and remove the top securing nut.
- Remove the retainer assembly and spring.
- Remove the securing nut of the shock absorber to remove the shock absorber. (5 × 1) (5)

1.3
- Worn upper ball joint
- Worn lower ball joint
- Collapsed coil spring
- Leaking shock absorber
- Worn control arm bushes
- Worn strut mountings (Any 3 × 1) (3)

1.4
- The tyre is damaged.
- The tyre is badly worn.
- The tyre is not running true.
- The rim is not running true.
- The rim is damaged. (Any 4 × 1) (4)

1.5
- Static balancing
- Dynamic balancing (2 x 1) (2)

1.6 Interlocking guard (1)

1.7
- Control panel
- Power switch
- Clamping nut
- Safety guard
- Wheel centralizing cones (Any 3 × 1) (3)

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QUESTION 2

2.1
- Remove the solenoid.
- Remove the through-bolts.
- Remove the front and rear housings.
- Remove the field coil frame.
- Remove the Bendix/starter clutch. (Any 5 × 1) (5)

2.2
- Pull-in test
- Hold-in test
- Pinion return test
- Pinion clearance check
- No-load test (Any 3 × 1) (3)

2.3 2.3.1 Setting the point gap (1)

2.3.2 The dwell angle (1)

2.3.3
- Loose base plate
- Worn base plate
- Worn distributor shaft or bushes
- Loose contact breaker (Any 2 × 1) (2)

2.3.4 A – Condenser
B – Distributor shaft
C – Contact breaker/Points (3)

2.3.5 Feeler gauge (1)

2.4
- The rotor
- The stator (2 × 1) (2)

2.5 2.5.1 The rectifier

2.5.2 The regulator (2 × 1) (2)

2.6
- Battery
- Ignition switch
- Alternator
- Warning lamp
- Wiring (Any 4 × 1) (4)

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QUESTION 3

3.1 3.1.1 To protect the fuel from the harmful effects of certain metals

3.1.2 To prevent the fuel from icing or freezing in the system

3.1.3 To keep the parts of the fuel system clean

(3 × 1) (3)

3.2 A stoichiometric mixture is the perfect air/fuel ratio of 14,9 parts of air to 1 part of fuel. (1 × 2) (2)

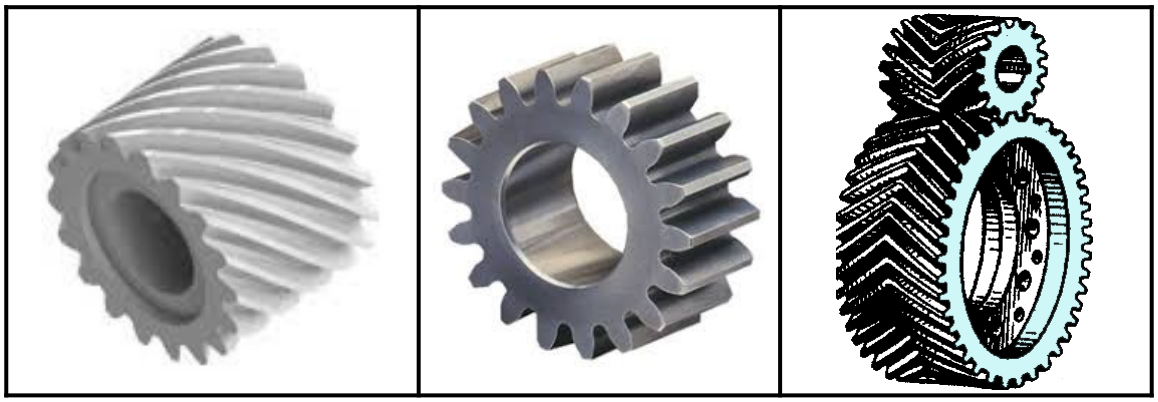
3.3
- Faulty fuel pump
- Blocked fuel filter
- Kinked fuel lines
- Incorrect float adjustment
- Sticky needle and seat (Any 3 × 1) (3)

3.4 Diesel does not combust easily in cold temperatures. Glow plugs are used to heat up the combustion chamber for easier cold start and smooth running. (2)

3.5 The brake force is dependent on the diameter of the cylinders. If the diameter of the wheel cylinder increase, the brake force also increases. (3)

3.6 A – Spring
B – Wheel cylinder piston
C – Cup washer/wheel cylinder seal/washer
D – Brake pipe or bleed nipple connection
E – Dust cover (5 × 1) (5)

3.7 3.7.1 3.7.2 3.7.3



Helical gear **Spur gear** **Double helical gear**

(3 × 2) (6)

3.8
- Manual bleeding/pumping the pedal
- Pressure bleeding/brake pressure bleeder (2)

3.9
- Crown wheel radial runout
- Crown wheel lateral runout
- Backlash
- Pinion pre-load
- Total pre-load
- Bolt torque specifications (Any appropriate 2 × 1) (2)

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QUESTION 4

4.1
- Worn/damaged piston
- Worn piston rings
- Worn cylinder (3)

4.2
- You need to know what the physical condition of the parts is before you send it away.
- Some damages can be hidden by dirt and oil.
- Cleaning the part will expose the damages or hidden cracks.

(Any 1 × 2) (2)

4.3 4.3.1 The camshaft should be replaced because it is responsible for the duration and the distance that the valve opens.

4.3.2 The valves should be checked and machined together with the valve seats in order to seal properly.

4.3.3 The piston should be replaced with another of the same size if the damage in the cylinder is minor. If not, the cylinders must be re-bored and new pistons fitted.

4.3.4 The crankshaft should be machined and new bearings must be fitted according to the machining.

(4 × 2) (8)

4.4
- Make sure the battery is fully charged or connect a charger.
- Run the engine to warm it up.
- Note the firing order and remove the plug leads.
- Clean around the spark plugs so that nothing falls in when removing the plugs.
- Remove the spark plugs and crank the engine to rid the cylinders of any carbon deposits that can get stuck between valve and seat. (2)

4.5
- A high-pressure cleaner
- A high-pressure cleaning machine for loose parts
- A solvent gun used with compressed air
- A firm paint brush used with solvent
- A parts cleaner/machine (Any 3 × 1) (3)

4.6
- A two-post hoist
- An engine hoist
- An engine stand
- A container to drain the oil
- Air-conditioner plant to remove the gas (Any 4 × 1) (4)

4.7
- A workshop manual provides the procedure that must be followed for the task.
- It helps to know in advance what the job entails.
- This means you can save time by not taking off unnecessary parts.
- All safety aspects of the task will be covered in the manual (Any 3 × 1) (3)

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TOTAL: 100

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